A synthesised Kerr nonlinearity: cascaded $\chi^{(2)}{:}\chi^{(2)}$ in $LiNbO_3$ metasurfaces

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Motivation and idea

Why we want a larger Kerr response

Kerr ($\chi^{(3)}$) underpins all-optical photonics: phase control, switching, four-wave mixing, frequency combs, all driven by a single pump.

Host	$\lvert\chi^{(2)}\rvert$	$\lvert\chi^{(3)}\rvert$ [m^2/V^2]	limitation
SiO_2	0	$\sim 3 \times 10^{-22}$	very weak Kerr
$LiNbO_3$	**50 pm/V**	$\sim 2 \times 10^{-21}$	**weak Kerr**
Si	0	$\sim 10^{-19}$	TPA + free carriers
AlGaAs	100 pm/V	$\sim 10^{-18}$	band-edge loss, heat

Materials with a *large* Kerr pay for it in loss and heat, while the cleanest $\chi^{(2)}$ platform, $LiNbO_3$, has a Kerr $\sim$100$\times$ *weaker* than silicon.

The same trade extends to the III–V semiconductors in our program, with engineered $\chi^{(2)}$ up to $\sim$10$\times$ $LiNbO_3$'s.

Ramesh *et al.*, Optica, in press; arXiv:2602.23246

The question

Can we *synthesise* a large, controllable $\chi^{(3)}$ from a strong $\chi^{(2)}$, instead of accepting a weak native Kerr?

Cascaded nonlinearities: prior work and what is new here

Two sequential $\chi^{(2)}$ processes, SHG followed by DFG: the round trip acts as an effective Kerr response.

- **1967**: predicted, as a nonlinear phase from phase-mismatched SHG. Ostrovskii, JETP Lett. 5, 272
- **1992**: measured, as a Z-scan through the phase-match angle in KTP; the nonlinear refraction *changes sign* at $\Delta k = 0$. DeSalvo *et al.*, Opt. Lett. 17, 28
- **1996–97**: reviewed; understood microscopically as a collective, virtual-SH effect. Stegeman *et al.*, Opt. Quantum Electron. 28, 1691; Khurgin *et al.*, JOSA B 14, 1977
- **2022**: cascading reaches metasurfaces, but for *harmonic generation* (THG via SHG+SFG in GaAs), not for a Kerr response on the pump. Gennaro *et al.*, ACS Photonics 9, 1026

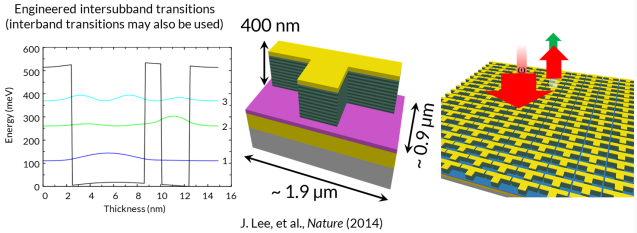
What is new here

In bulk, the cascade lever is the phase mismatch: $\chi^{(3)}_{\mathrm{casc}} \propto 1/\Delta k$. We replace it with a **resonant lever**, a high-Q, *dark* resonance at the second harmonic: $\chi^{(3)}_{\mathrm{casc}} \propto Q_2/\Delta\omega$, detuning-tunable, chip-scale, and CW-compatible.

Our starting point: quantum-engineered $\chi^{(2)}$ metasurfaces

Quantum-engineered semiconductor heterostructures
+
Electromagnetically-engineered modes in metallic nanostructures

Large-area ultrathin metasurfaces with tailored nonlinear response

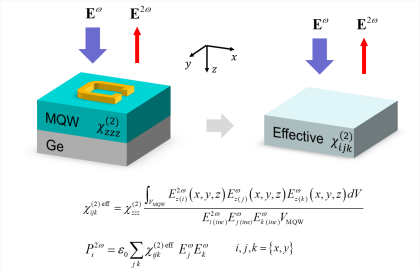


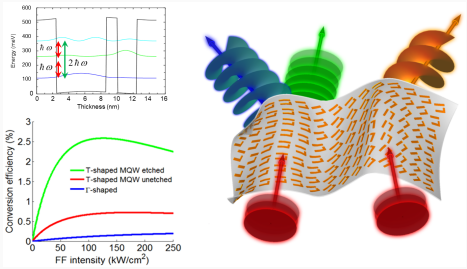
J. Lee, et al., *Nature* (2014)

Intersubband transitions in MQW heterostructures coupled to plasmonic modes: a 400-nm-thick metasurface with $|\chi^{(2)}| > 5 \times 10^4$ pm/V at $\lambda \approx 8\,\mu$m, SHG at normal incidence.

J. Lee *et al.*, Nature 511, 65 (2014)

From giant $\chi^{(2)}$ to full control of the second harmonic



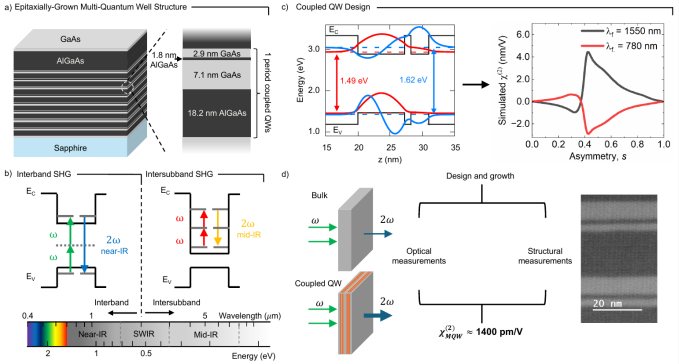


Homogenised effective $\chi^{(2)}_{ijk}$ per meta-atom; SHG conversion up to $\sim$2.5 %, and gradient (Pancharatnam–Berry) phase control of the emitted second harmonic. Tymchenko *et al.*, PRL 115, 207403 (2015); Nookala *et al.*, Optica 3, 283 (2016)

This project

Use the engineered $\chi^{(2)}$ *twice*: convert it into an effective Kerr response on the pump.

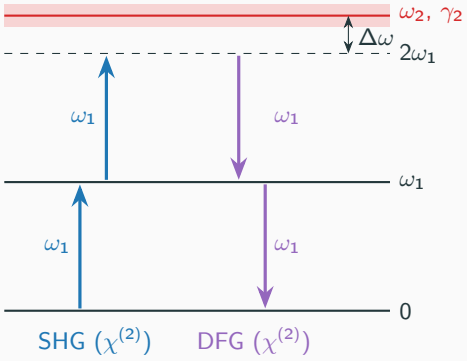
Giant interband $\chi^{(2)}$ from coupled QWs



Asymmetric GaAs/AlGaAs coupled quantum wells: measured interband $\chi^{(2)} \approx 1400\,\text{pm/V}$, up to 2750 pm/V, at $\lambda \approx 1.55\,\mu\text{m}$ ($\sim$10$\times$ $LiNbO_3$); designs predict up to $\sim$7 nm/V. A quantum-engineered, MBE-grown large-$\chi^{(2)}$ material for the cascade.

R. Ramesh *et al.*, "Enhanced Interband Optical Nonlinearities from Coupled Quantum Wells," Optica, in press; arXiv:2602.23246

The idea in one picture



The SH is **virtual**: detuned by $\Delta\omega = 2\omega_1 - \omega_2$ from a dark, high-Q_2 mode, it never radiates; it only back-acts on the pump. A second-order process through the SH Green's function,

$$\chi^{(3)}_{\mathrm{casc}} \;=\; \chi^{(2)} \cdot G(2\omega_1) \cdot \chi^{(2)}, \qquad G \;\propto\; \frac{\mathrm{f}_2(\mathrm{r})\,\mathrm{f}_2^{*}(\mathrm{r}')}{\Delta\omega + i\gamma_2/2},$$

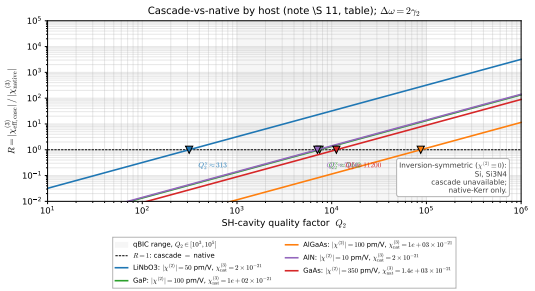
which, reduced to the single dark mode, gives

$$\boxed{\chi^{(3)}_{\mathrm{casc}}(\Delta\omega) = -\,\frac{\omega_1\,|\chi^{(2)}|^2\,\eta_{\mathrm{geom}}}{6\,n_2^2\,(\Delta\omega + i\gamma_2/2)}}$$

- Re: a Kerr response, $\propto 1/\Delta\omega$, **sign-tunable** (focusing $\leftrightarrow$ defocusing);
- Im: a two-photon-like loss, Lorentzian at $\Delta\omega = 0$: a *detunable* saturable absorber;
- η_{geom}: the $\chi^{(2)}$-weighted overlap of the two modes, the principal design parameter.

Figure of merit: enhancement over the native Kerr

$$R \equiv \frac{|\chi^{(3)}_{\mathrm{casc}}|}{|\chi^{(3)}_{\mathrm{native}}|}, \qquad \chi^{(3)}_{\mathrm{casc}}\Big|_{\Delta\omega=2\gamma_2} \simeq \frac{Q_2}{24\, n_2^2}\, |\chi^{(2)}|^2\, \eta_{\mathrm{geom}}$$



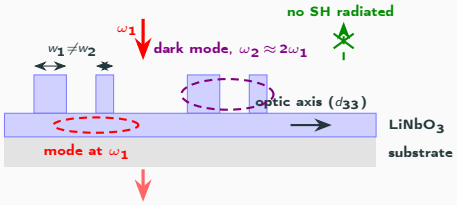
Host	R @ Q_2=4000	crossover Q_2
$LiNbO_3$	≈ 12	$\sim 3 \times 10^2$
GaP	0.5	$\sim 7 \times 10^3$
AlN	0.6	$\sim 7 \times 10^3$
AlGaAs	0.05	$\sim 9 \times 10^4$

$LiNbO_3$ is the favourable host *because* its native Kerr is weak (crossover $Q_2 \sim 300$); the framework extends to the materials in our program (outlook slide).

Uncertainty: absolute R carries an $O(1)$ prefactor uncertainty; scalings and ratios are robust.

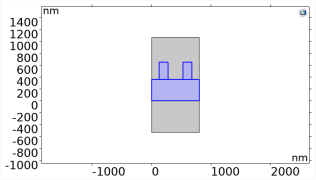
Platform and theory

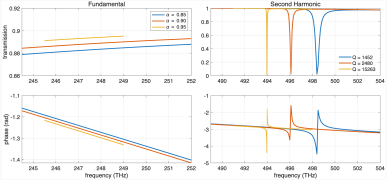
Platform: engineer the *second-harmonic* resonance



An inversion of our usual design philosophy: the resonance that does the work sits at the **SH**, not at the pump.

- symmetry-protected qBIC at $\omega_2 \approx 2\omega_1$: high Q_2, *dark* (no SH radiates); meta-atom asymmetry = the Q_2 knob;
- $LiNbO_3$ thin film, x-cut: d_{33} with $E \parallel$ optic axis; fabrication constraints already in the designs;
- pump non-resonant (**singly resonant, 1R**) or sharing a qBIC (**doubly resonant, 2R**).





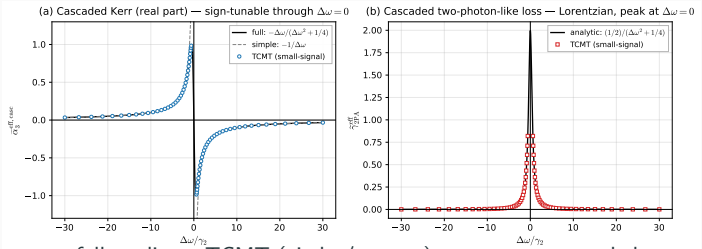
Unit cell (COMSOL); FM band featureless, SH qBIC Q tuned 10^3–10^4 by the asymmetry.

Theory in two lines (coupled-mode equations)

$$\frac{da_2}{dt} = \left(i\Delta\omega - \frac{\gamma_2}{2}\right) a_2 + i\beta\, a_1^2, \qquad \frac{da_1}{dt} = -\frac{\gamma_1}{2}\, a_1 + i\beta^* a_1^* a_2 + i\alpha_3 |a_1|^2 a_1 + \sqrt{\gamma_1^{\mathrm{rad}}}\, s_+$$

The *same* β in SHG and DFG terms $\Rightarrow$ Manley–Rowe. Adiabatic elimination of a_2 (valid for $|\Delta\omega| \gtrsim 2\gamma_2$):

$$\alpha_3^{\mathrm{casc}}(\Delta\omega) = -\frac{|\beta|^2\, \Delta\omega}{\Delta\omega^2 + (\gamma_2/2)^2} \quad \text{(Kerr)}, \qquad \gamma_3^{\mathrm{casc}}(\Delta\omega) = \frac{|\beta|^2\, \gamma_2/2}{\Delta\omega^2 + (\gamma_2/2)^2} \quad \text{(TPA-like)}$$



Closed forms vs full nonlinear TCMT (circles/squares): exact agreement below saturation.

Best case: the doubly-resonant metasurface

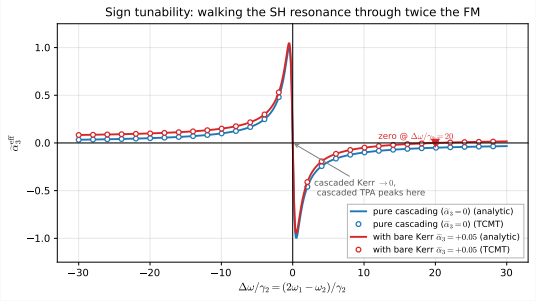
Leverage on Kerr observables, and new capabilities

With the pump also resonant (Q_1), every Kerr observable inherits the cascade:

Observable	gain vs native
SPM / XPM phase	$1 + R \approx 13$
Bistability threshold	$\frac{1}{1+R} \approx \frac{1}{13}$
Switching energy	$\frac{1}{1+R}$
FWM efficiency	$(1 + \boldsymbol{R})^2 \approx 170$

FWM is the most differentiating observable (TCMT-validated, backup).

Cascade-only	key property
Saturable absorber	TPA peak $\propto Q_2$, detunable, ps
Passive limiter	clamp at $\lvert\beta a_1\rvert = \lvert\Delta\omega\rvert$
$\Delta\omega$-readout sensor	slope $\propto Q_2^2$ ($Q_2^{3/2}$ shot-noise)



Sign control, unavailable in native Kerr: self-focusing $\leftrightarrow$ defocusing in one device (thermo-optic / Pockels tuning of ω_2).

Limitations and trade-offs

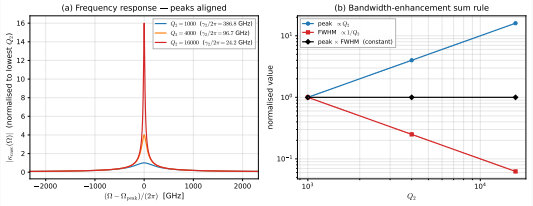
Four intrinsic limitations

1. **Cascaded TPA.** Loss/Kerr $= \gamma_2/(2|\Delta\omega|)$: 1/4 at the conservative $\Delta\omega = 2\gamma_2$; a loss channel absent in native $LiNbO_3$.

2. **Sum rule.** $|\chi^{(3)}_{\mathrm{casc}}|_{\mathrm{peak}} \times \Delta f$ is fixed: Q_2 only reshapes the Lorentzian. Bandwidth $\approx 100\,\mathrm{GHz}$ at $Q_2 = 4000$; ultrafast operation is excluded.

3. **Saturation ceiling.** Parametric instability at $|\beta a_1| = |\Delta\omega|$: usable pump window ≈ 1 decade at $Q_2 = 4000$, shrinking as $1/Q_2^2$.

4. **Speed** $\sim 1/\gamma_2 \sim$ ps: fine for CW/GHz, closed to fs applications.



Peak $\propto Q_2$, FWHM $\propto 1/Q_2$, product constant: you cannot beat the sum rule by pushing Q_2.

Fabrication constraints: the case for starting singly-resonant

The doubly-resonant device needs: two co-localised high-Q modes, ω_2 aligned to $2\omega_1$ against material dispersion, a tensor-correct overlap, *and* array uniformity:

Q_2^{array} collapses once $\sigma_\omega/\omega \gtrsim 1/Q_2$ (sub-nm uniformity of the symmetry-breaking feature).

Design study (full-wave, $LiNbO_3$)

- **Doubly resonant:** $Q_2 \approx 5000$ achievable with $\sim$10 % overlap; mode alignment is the main difficulty.
- **Singly resonant:** $Q_2 \approx 8000$ with $\sim$20 % overlap; substantially easier, tunable by asymmetry alone.

$\Rightarrow$ **strategy:** demonstrate and calibrate the cascade on the *singly-resonant* platform first; develop the doubly-resonant design in parallel for the eventual Kerr device.

The singly-resonant implementation

What survives with only the SH resonance

	doubly resonant (2R)	SH-only (1R)
$\chi^{(3)}_{\mathrm{casc}}$, same boxed form	yes	yes
Sign flip in $\Delta\omega$ (the fingerprint)	yes	yes
Detunable TPA / saturable loss	yes	yes
$R > 1$ (beat the native Kerr)	yes	yes (if overlap is good)
Pump-cavity leverage $\sim Q_1^2$ on φ_{NL}	yes	no (single pass: μrad)
Bistability, parametric ceiling, CW device	yes	no

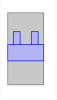
Role of the 1R experiment

Not a device, but a **materials-level demonstration and calibration** of $\chi^{(3)}_{\mathrm{casc}}$: measure the sign flip, extract the magnitude, validate the model that the 2R device design rests on.

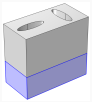
Five fabricable singly-resonant designs

	λ_1	Q_2	overlap	R	>1?
S1	997 nm	3442	0.12	0.49	**no**
S2	975 nm	3486	0.11	0.39	**no**
S3	1058 nm	4641	0.19	1.76	**yes**
S4	1064 nm	5633	0.21	**2.49**	**yes**
D6	**1757 nm**	**924**	**0.43**	**1.81**	**yes**

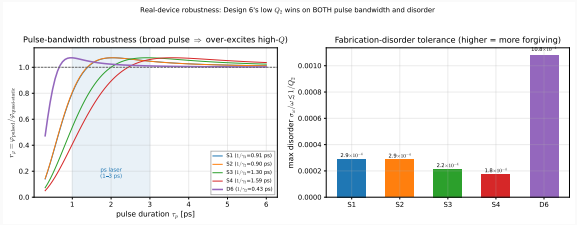
S1–S4: pure-$LiNbO_3$ pillars (height asymmetry); **D6**: TiO_2/$LiNbO_3$ hybrid (etch-angle asymmetry). Operating point $\Delta\omega = +2\gamma_2$; $R \propto Q_2 \times \mathrm{overlap}^2$.



S1–S4



D6



A 1–3 ps pulse is spectrally broad: high-Q designs generate up to **22 %** of the SH at the *wrong* sign of $\Delta\omega$ (cancellation), so S4's Kerr phase drops to 0.4× at 1 ps. **D6** is unaffected ($r_\varphi \approx 1$) and tolerates 5× more fabrication disorder.

Best in the ideal limit: S4 ($R = 2.49$).
Best for a real device: D6 ($R \approx 1.8$, robust).

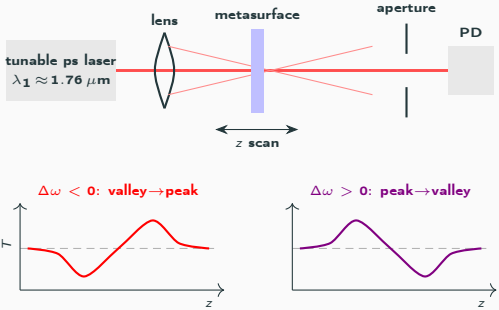
Proposed experiment: a resonant analogue of the 1992 Z-scan

Closed-aperture Z-scan vs pump detuning: the DeSalvo (1992) KTP measurement with the phase-match angle replaced by a *resonance* detuning.

- tune the pump by a few nm $\Rightarrow$ the generated $2\omega_1$ sweeps across the SH mode, $\Delta\omega : - \to +$;
- valley–peak $\to$ peak–valley reversal at $\Delta\omega = 0$: the cascade is *odd* in $\Delta\omega$, native Kerr a constant background; $R > 1 \Rightarrow$ the *total* n_2 reverses sign;
- SHG efficiency on the same sample pins the absolute $\chi^{(3)}$ scale.

Vehicle: Design 6 at $\lambda_1 \approx 1.76\,\mu$m

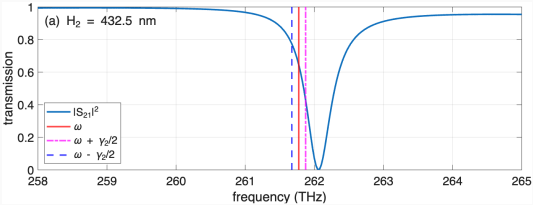
ps laser reaches it; SH lifetime 0.43 ps $\Rightarrow$ quasi-static at $\tau_p = 1$–3 ps; sign flip stays sharp.



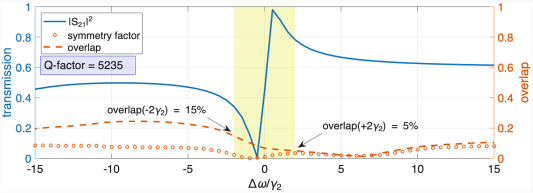
The closed-aperture trace *mirrors* as the pump crosses $\Delta\omega = 0$: the cascade's fingerprint. Predicted spectra: backup.

Outlook: the doubly-resonant device and new materials

The doubly-resonant path: a first candidate

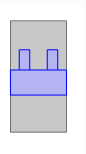


Pump resonance at ω_1; markers show the operating pump frequency and $\omega \pm \gamma_2/2$ inside the line.



SH qBIC: $Q_2 = 5235$, overlap 15 % at $\Delta\omega = -2\gamma_2$.

- Full-wave design with **co-localised pump and SH modes**: the central difficulty of the 2R scheme is already solved at the design level.
- $Q_2 \times \mathrm{overlap}^2$ comparable to the best singly-resonant samples, and the pump cavity restores the device-level leverage ($1{+}R$ phase, bistability, FWM).
- Observable-level analysis of this design is the next iteration; the 1R calibration fixes its absolute scale.



LN pillar-pair cell

Beyond $LiNbO_3$: MURI material platforms

$LiNbO_3$ is the calibration baseline; the formalism is material-agnostic: $R \propto |\chi^{(2)}|^2\, Q_2\, \eta_{\mathrm{geom}} \,/\, \chi^{(3)}_{\mathrm{native}}$.

- **III–V semiconductors (GaAs, GaP; MBE growth within the team).** Bulk $|\chi^{(2)}|$ up to $\sim$350 pm/V; team-demonstrated interband coupled-QW $\chi^{(2)}$ up to 2750 pm/V at 1.55 μm (Ramesh *et al.*, Optica, in press; arXiv:2602.23246). At telecom the band-edge native Kerr dominates ($R < 1$); pumping **below half the gap** (mid-IR) suppresses the native Kerr by roughly an order of magnitude and removes intrinsic TPA ($2\hbar\omega < E_g$) $\Rightarrow R > 1$ at the same Q_2, with the SH still below the gap.
- **MQW intersubband metasurfaces** (our Nature 2014 platform): engineered $|\chi^{(2)}| \sim 5 \times 10^4$ pm/V at $\lambda \approx 8\,\mu$m; a natural **next step** for the cascade. The same quantum engineering must also *suppress* the (equally giant) native intersubband $\chi^{(3)}$; SH-band absorption sets Q_2.
- **Epitaxial $BaTiO_3$ on Si** (team growth): optical d coefficients below $LiNbO_3$'s, but a giant Pockels response (r_{eff} up to $\sim$430 pm/V): exactly the $\Delta\omega$-tuning knob the cascade needs (sign flip, AM modulation), monolithic on Si. Demkov group, JVST A 39, 030804 (2021)

Honest summary

Honest summary

What is real today

- $R \approx 1.8$–2.5 over native Kerr in fabricable single-resonance $LiNbO_3$ designs;
- a clean, unambiguous fingerprint: the **sign flip**, odd in $\Delta\omega$;
- a robust vehicle (D6): immune to pulse bandwidth and $5\times$ looser on disorder;
- validated theory: closed form = full TCMT.

What is not (yet)

- a CW Kerr *device*: single-pass phase is μrad; this requires the doubly-resonant version ($1+R$ leverage, $R \sim 12$);
- absolute R: order-of-magnitude until the SHG calibration pins the overlap normalisation;
- bandwidth beyond $\sim$100 GHz: the sum rule forbids it.

Next steps

(1) ps Z-scan sign-flip on Design 6 at 1.76 μm + SHG calibration (pins the absolute scale). **(2)** Advance the doubly-resonant candidate: the device-level payoff. **(3)** Realise the cascade on the MQW intersubband platform: quantum-engineered $\chi^{(2)}$ with the photonic $Q_2/\Delta\omega$ lever.

Backup

Backup: adiabatic elimination, explicitly

Setting $\dot{a}_2 \to 0$ for $|\Delta\omega| \gtrsim 2\gamma_2$: $\quad a_2 = -\beta\, a_1^2/(\Delta\omega + i\gamma_2/2)$, so that

$$\frac{da_1}{dt} = -\tfrac{\gamma_1}{2} a_1 + i\Big[\alpha_3 - \frac{|\beta|^2\Delta\omega}{\Delta\omega^2 + (\gamma_2/2)^2}\Big]|a_1|^2 a_1 - \frac{|\beta|^2\gamma_2/2}{\Delta\omega^2 + (\gamma_2/2)^2}\,|a_1|^2 a_1 + \sqrt{\gamma_1^{\mathrm{rad}}}\, s_+$$

$$\beta = \frac{\varepsilon_0\omega_1}{4}\int \chi^{(2)}_{ijk}(\mathrm{r})\, f^*_{2,i} f_{1,j} f_{1,k}\, d^3r, \qquad \alpha_3 = \frac{3\varepsilon_0\omega_1}{8}\int \chi^{(3)}_{ijkl} f^*_{1,i} f_{1,j} f_{1,k} f^*_{1,l}\, d^3r$$

Modes are energy-normalised (Brillouin form $\widetilde{\varepsilon} = \partial(\omega\varepsilon_r)/\partial\omega$); $|a_n|^2$ = stored energy. Boyd degeneracy factors (SHG 1, DFG 2, SPM 3) are what make the same β appear in both equations (Manley–Rowe).

SH-only case: the same elimination with the local (non-resonant) FM field gives

$$\chi^{(3),1\mathrm{R}}_{\mathrm{casc}} = -\frac{\varepsilon_0\,\omega_1\,|\widetilde{M}|^2}{6\,(\Delta\omega + i\gamma_2/2)\,V_{\mathrm{cell}}}, \qquad \eta^{1\mathrm{R}}_{\mathrm{geom}} = \big(|\mathrm{overlap}|/V\big)^2.$$

Backup: $\chi^{(3)}_{\mathrm{casc}}$ from the polarisation chain alone

No mode decomposition; three steps at the level of macroscopic polarisations (Boyd convention, $\mathcal{E} = \frac{1}{2}[E e^{-i\omega t} + \mathrm{c.c.}]$):

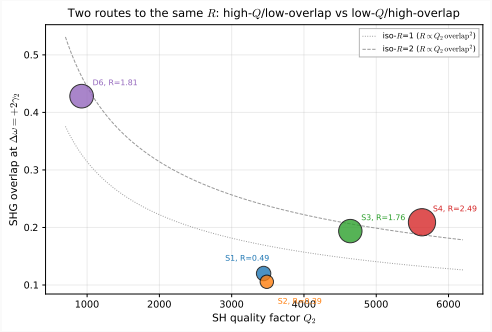
1. **SHG drive:** $P^{(2)}(2\omega_1) = \varepsilon_0\, \chi^{(2)}\, E(\omega_1)^2$ (single ordering, no degeneracy factor)
2. **Propagation:** $E(2\omega_1; \mathrm{r}) = \mu_0 (2\omega_1)^2 \int G(\mathrm{r}, \mathrm{r}'; 2\omega_1)\, P^{(2)}(\mathrm{r}')\, d^3 r'$, ($G$: Green's function of the wave operator at $2\omega_1$)
3. **DFG return:** $P^{(3)}(\omega_1) = 2\,\varepsilon_0\, \chi^{(2)}\, E(2\omega_1)\, E^*(\omega_1)$ (two orderings, Boyd factor 2)

Identifying with the SPM form $P^{(3)}(\omega_1) \equiv 3\varepsilon_0\, \chi^{(3)}_{\mathrm{casc}}\, |E|^2 E$: $\chi^{(3)}_{\mathrm{casc}} = \frac{2}{3}\, \mu_0 (2\omega_1)^2\, \chi^{(2)} \cdot G(2\omega_1) \cdot \chi^{(2)}$ (schematic). Two limits of G recover both regimes:

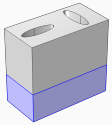
- **bulk SVEA** (phase-mismatch kernel): $\chi^{(3)}_{\mathrm{casc,bulk}} = -\dfrac{2\omega_1\, |\chi^{(2)}|^2 F}{3\, c\, n_2\, \Delta k}$, the PPLN regime;
- **single resonant mode** (QNM expansion): $G \to \dfrac{c^2}{2\omega_2}\, \dfrac{\mathrm{f}_2(\mathrm{r})\mathrm{f}_2^*(\mathrm{r}')}{\Delta\omega + i\gamma_2/2}$, giving the boxed metasurface result.

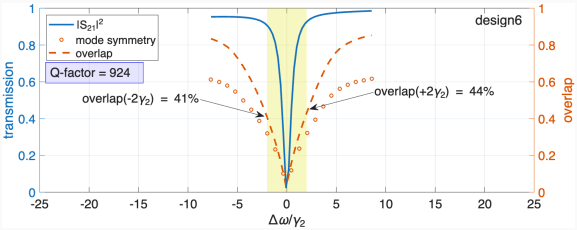
Sauvan *et al.*, PRL 2013; Lalanne *et al.*, LPR 2018

Backup: two routes to $R \approx 2$; Design 6 spectrum



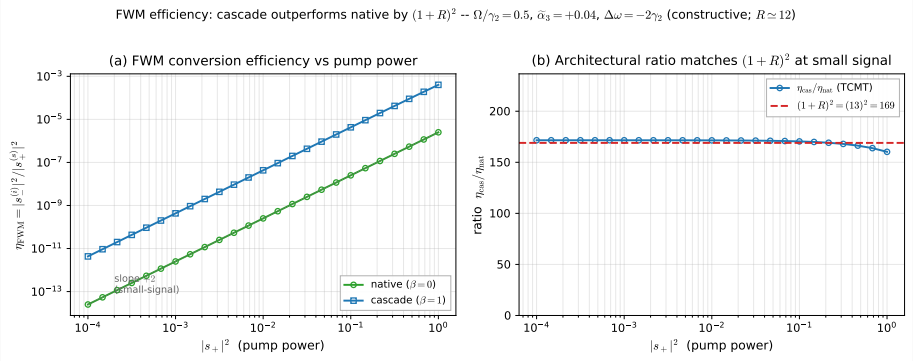
Iso-R curves $R \propto Q_2 \mathrm{overlap}^2$: S4 (high Q, low overlap) and D6 (low Q, high overlap) reach comparable R.





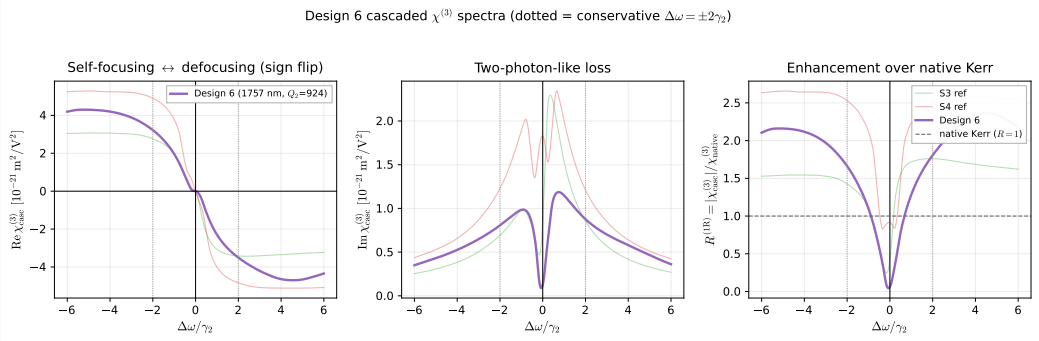
Design 6: $Q_2 = 924$, overlap 0.41/0.44 at $\mp 2\gamma_2$.

Backup: FWM enhancement $(1+R)^2$, TCMT-validated



Pump + weak signal inside the fundamental linewidth; idler efficiency scales as $|\alpha_3^{\rm eff}|^2$, so the architectural ratio is $(1+R)^2 = 169$ at the $LiNbO_3$ design point. Validity: $\Omega < \min(\gamma_1, \gamma_2)$, small-signal cascade.

Backup: Design 6 predicted $\chi^{(3)}$ spectra



Sign-tunable $\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}}$, Lorentzian-like $\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}$ (two-photon-like loss), and R up to $\sim$2.2–2.4 in the wings; dotted lines mark the conservative $\Delta\omega = \pm 2\gamma_2$ operating points.

Backup: host-material survey

Host	$\lvert\chi^{(2)}\rvert$ [m/V]	$\lvert\chi^{(3)}_{\mathrm{native}}\rvert$ [m^2/V^2]	n_2	R @ Q_2=4000	Q_2^{cross}
$LiNbO_3$	5×10^{-11}	2×10^{-21}	2.26	**12.8**	313
GaP	1×10^{-10}	1×10^{-19}	3.05	0.54	7 442
AlGaAs	1×10^{-10}	1×10^{-18}	3.30	0.05	87 120
AlN	1×10^{-11}	2×10^{-21}	2.10	0.57	7 060
GaAs	3.5×10^{-10}	1.4×10^{-18}	3.50	0.36	11 200
Si, Si_3N_4	0	—	—	— (no cascade)	—

Conservative operating point $\Delta\omega = 2\gamma_2$, $\eta_{\mathrm{geom}} = 0.3$ assumed for all hosts. Semiconductors lose because their band-edge-enhanced *native* Kerr is even larger than their $|\chi^{(2)}|^2$ advantage.

Backup: dynamic range vs Q_2

Floor: readable phase $\varphi_{\mathrm{NL}} \gtrsim 10\,\mathrm{mrad}$. Ceiling: parametric instability $|\beta a_1| = |\Delta\omega|$, $|a_1|^2_{\mathrm{sat}} \propto 1/Q_2^2$ at fixed $\Delta\omega/\gamma_2$.

Q_2	R	usable pump window	verdict
1 000	≈ 3	$\sim$3 decades	disorder-tolerant, modest gain
4 000	≈ 12	$\sim$1 decade	**design point** (already at the edge)
10 000	≈ 30	$\sim$0.3 decade	marginal
10^5	≈ 300	$\sim$0.03 decade	unusable

Want more leverage *and* range? Increase $|\chi^{(2)}|$ or η_{geom}, not Q_2.

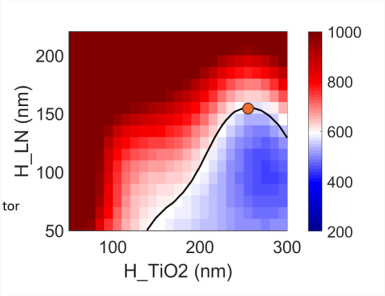
Backup: disorder / inhomogeneous broadening

qBIC arrays average over meta-atoms: the array linewidth collapses once $\sigma_\omega/\omega \gtrsim 1/Q_2$ (equivalently $\sigma_\alpha/\alpha \lesssim 1\%$ for the headline 2R point).

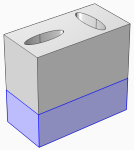
	S1	S2	S3	S4	D6
max σ_ω/ω	2.9×10^{-4}	2.9×10^{-4}	2.2×10^{-4}	1.8×10^{-4}	1.1×10^{-3}

Mitigations for high-Q routes: smaller pump spot / single-shot e-beam field, post-fabrication trimming, thermo-optic feedback, ensemble-averaged readout.

Backup: Design 6 details

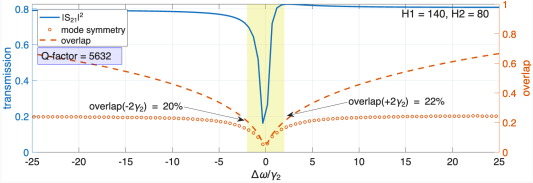


Q_2 over (H_{TiO_2}, H_{LN}); chosen point marked.

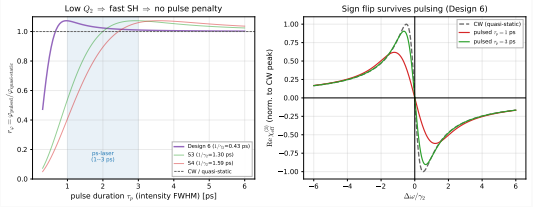


- TiO_2 pillars (two elliptical holes, etch angles $0°/90°$) on a *solid* $LiNbO_3$ slab; only the LN is nonlinear.
- $H_{LN} = 150\,\text{nm}$, $H_{TiO_2} = 246\,\text{nm}$, axes $176 \times 58\,\text{nm}$, period $219\,\text{nm}$ ($2p \times p$ supercell).
- Optic axis in-plane ($\| x$): E_x input, d_{33} channel, $n_1 = 2.13$, $n_2 = 2.17$ (n_e); effective hybrid index within 1 % of bare LN.
- Verified: Fano fit $Q = 920$ vs quoted 924; volume $= 2p^2 H_{LN}$ exact; overlap 0.41/0.44 reproduced; FM transmission flat (0.98).
- Optimal detuning: R peaks ≈ 2.36 at $\Delta\omega \approx +4.5\gamma_2$ (overlap keeps rising in the wings).

Backup: sample spectra (S4) and pulsed response

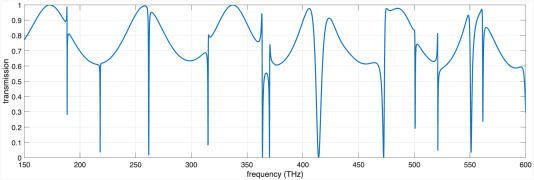


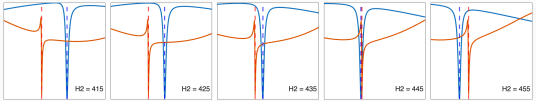
S4: $Q_2 = 5633$, overlap 0.20/0.22 at $\mp 2\gamma_2$; the best design in the ideal limit.



D6 time-domain pulsed TCMT: quasi-static at 1–3 ps ($r_\varphi = 1.07/1.02/1.01$), sign flip survives.

Backup: doubly-resonant design progress





Full-wave search for co-localised qBIC pairs at $(\omega_1, 2\omega_1)$: mode alignment via geometry, overlap dominated by the symmetry-matched channel; $Q_2 \approx 5000$ with $\sim$10 % overlap currently achievable.